Intelligent UAS Knowledge Platform

Centralized Tools for Army Drone Operations

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Problem Statement

(1) Fragmented UAS data sources (spreadsheets, white papers, CAD/software repos).

(2) Time-consuming comparisons & reporting.

(3) No central place for lessons learned (AARs) & operator collaboration

Web-based hub for all UAS data.

Search & Compare Tool (filter & compare UAS)

UAS News Aggregation (auto-add articles by URL, admin moderation)

CAD/Software Library (admin-only uploads, user downloads)

AAR Submissions (with voting, comments, attachments)

Drone Team Portal (applications, roster, events calendar)

Profile & Recommendations (custom user experience)

Architecture Overview

Frontend: Flask + Jinja templates, modular nav.

Backend: Python (Flask Blueprints), services layer for UAS/AAR/News/Software.

Data: CSV-backed storage, static uploads, white paper DB.

Integrations:

- Email (mailer.py).
- Google Calendar embed.
- Metadata scraping for news ingestion.
- Auth: Role-based (admin vs user).

Key Benefits

- ✓ Centralized UAS data source.
- ✓ Speeds decision-making with fast comparisons.
- ✓ Promotes collaboration with AAR voting/comments.
- ✓ Secure: admin-only uploads & news moderation.
- ✓ Extendable: easy to add new modules/features.
- ✓ Future-ready: tie into MDMP

Vision & Future Growth

AI-driven analytics: auto-summarize AARs, recommend UAS for missions.

Integration with live ops data (flight hours, readiness).

Cloud deployment for broader access.

Potential extension into MDMP automation.